

Name : NANDANA JAYAN
Roll number : 25MML1016
Registered email address : nandanajayan13@gmail.com
Institution : VIT Chennai
Course : M.Tech in AI and ML

Individual Project 1

Project title. : Agentic AI-Based Construction Schedule Disruption
Analysis and Recovery Planning System

Summary :

This project proposes an Agentic AI system that autonomously analyzes the impact of unexpected construction schedule disruptions, such as task delays, worker shortages, material unavailability, or equipment failures, and generates feasible recovery plans. The system will analyze task dependencies, available resources, deadlines, and associated costs to identify affected activities and evaluate alternative recovery strategies. Using multiple specialized AI agents and optimization techniques, the system will dynamically select a cost-effective plan to minimize project delays and re-plan the schedule when conditions change. The solution will be implemented as a lightweight simulation-based system with an interactive dashboard, making it feasible to develop and deploy within the given timeframe without requiring GPU-intensive computation.

Individual Project 2

Project title. : An Explainable RAG Framework for Enterprise Policy
Change Impact Analysis

Summary :

This project proposes an Explainable Retrieval-Augmented Generation (RAG) system that analyzes the impact of changes made to enterprise policies by identifying related policies, dependent clauses, potential conflicts, and downstream effects. The system

will use RAG to retrieve relevant policy sections, extract and compare policy requirements, and construct a relationship graph connecting policies and their clauses. When a policy requirement is modified, the system will trace the affected policies and provide an evidence-grounded impact report, clearly showing the source documents and clauses responsible for each finding. The proposed solution will use lightweight RAG, graph-based relationship analysis, and an LLM for contextual interpretation and explanation, making it feasible to develop and deploy within two months on a CPU-based company laptop.

Group Project :

Team Members

With roll No :Nandana Jayan(25MML1016)

Sanika Yashwant Gaikwad(25MML1004)

Anuja (25MML1026)

Project title. :AI-Powered Root Cause and Decision Intelligence System

Summary :

This project proposes an AI-powered decision intelligence system that goes beyond detecting anomalies by investigating why an event occurred, what evidence supports the identified root cause, what its potential impact could be, and what actions should be taken. The system will analyze diverse sources such as logs, sensor data, KPIs, historical events, documents, and business data to identify anomalies, correlate relevant evidence, determine probable root causes, assess their potential impact, and generate predictive and actionable recommendations. Machine learning and statistical techniques will provide the underlying analytical evidence, while Generative AI will be used for reasoning, evidence-based explanation, and decision support. The framework can be adapted to domains such as IT operations, manufacturing, supply chain, energy, and business operations, making it a flexible and practical AI solution.